

(a)

PRIOR STATE

state belief $b_{t-1}(s)$

Emotion disclosure 0.30

Emotion processing 0.45

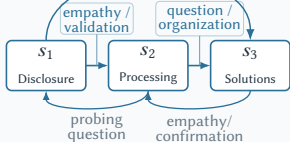
Considering solutions 0.25

(b)

PREDICT

transition model
 $P(s_t | s_{t-1})$

premature advice



$$\hat{b}_t(s_t) = \sum_{s_{t-1} \in S} P(s_t | s_{t-1}) b_{t-1}(s_{t-1})$$

predicted distribution $\hat{b}_t(s_t)$

(c)

LABEL RESPONSE

"That sounds
really difficult."candidate
response r_t

LLM

observed label o_t
Empathy / validation

(d)

BAYESIAN UPDATE

observation o_t prediction $\hat{b}_t$

$$b_t(s_t) = \frac{P(o_t | s_t) \hat{b}_t(s_t)}{\sum_{s' \in S} P(o_t | s') \hat{b}_t(s')}$$

posterior belief $b_t(s) = P(s_t | o_{1:t})$

Disclosure 0.12

Processing 0.66

Solutions 0.22

(e)

PROGRESSION SCORE

$$\text{Score}(r_t) = \sum_{s \in S_{\text{target}}} b_t(s)$$

 $S_{\text{target}} = \{\text{Processing, Solutions}\}$
 $b_t(\text{Processing}) + b_t(\text{Solutions})$ $= 0.66 + 0.22$ **Score(r_t) = 0.88**

progression compatibility